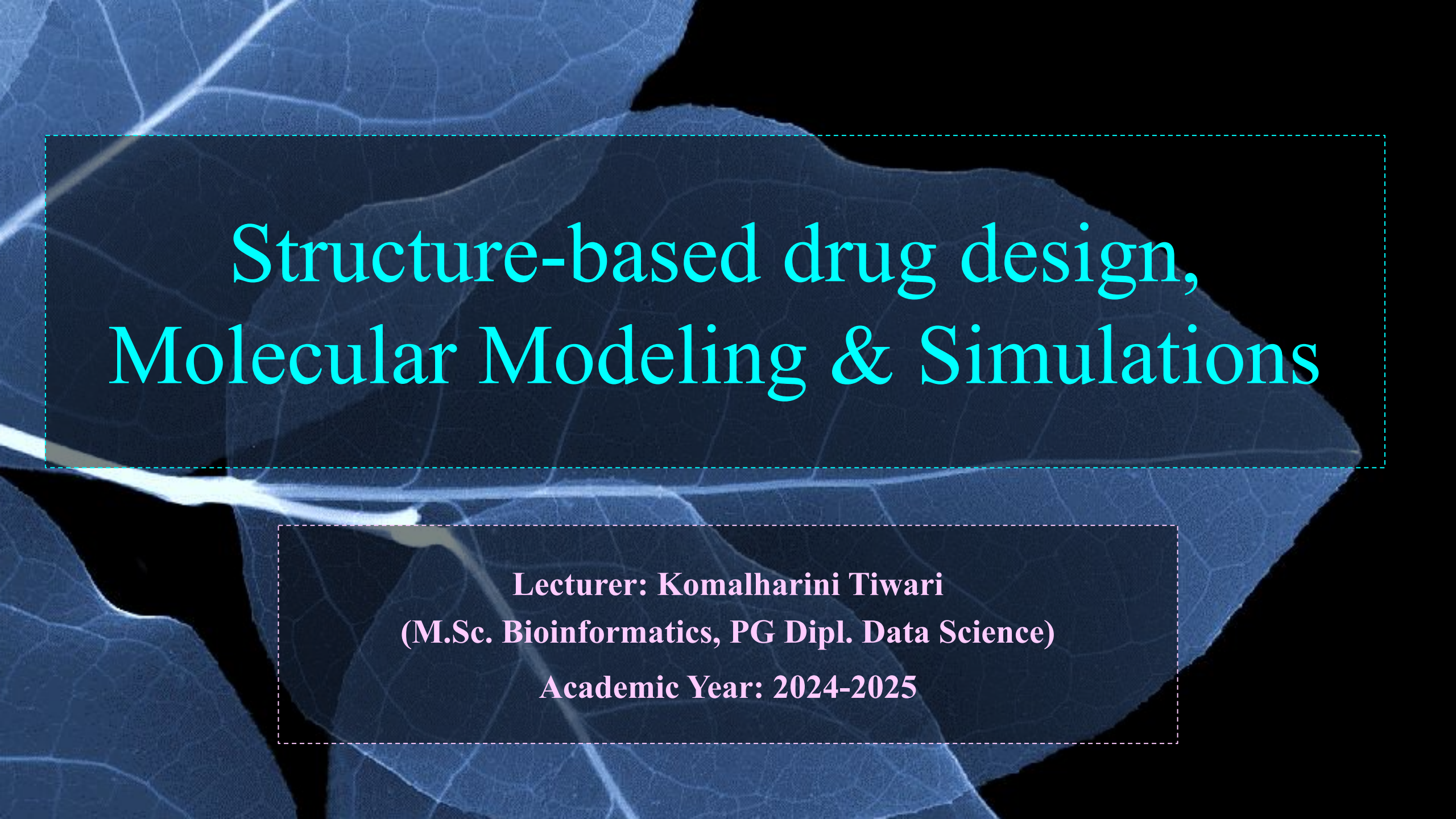




Structure-based drug design, Molecular Modeling & Simulations

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Academic Year: 2024-2025

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4. Molecular Dynamics: Newton's equation of motion, equilibrium point, radial distribution function, pair correlation functions, duration of MD run

Introduction to Molecular Docking:

- ❑ Molecular docking is like trying to fit a key (drug molecule) into a lock (protein pocket). The goal is to predict if the key fits well and opens the lock effectively.
- ❑ The interaction between the key and the lock depends on how the molecules' shapes and forces match.
- ❑ The goal is to understand the binding mode, strength, and orientation of the ligand in the active site of the receptor. This is commonly used in drug discovery to identify molecules that can inhibit or activate a target.
- ❑ Molecular docking involves two components:
 - ▶ Ligand: A small molecule (like a drug).
 - ▶ Receptor: The target molecule, usually a protein.
- ❑ Types of Docking:
 - ▶ Rigid Docking: Both receptor and ligand structures remain fixed.
 - ▶ Flexible Docking: Allows movement in the ligand or specific receptor parts.

Introduction to Molecular Docking:

- Docking predicts how a ligand will bind to a receptor's active site. It determines:
 - ▶ Binding orientation: How the ligand fits.
 - ▶ Binding affinity: Strength of interaction.
- Steps in Docking:
 - ▶ Preparation: Clean the protein and ligand structures.
 - ▶ Docking Algorithm: Use software (e.g., AutoDock, Vina) to predict the pose of the ligand.
 - ▶ Validation: Compare docking results with experimental data to ensure accuracy.
 - ▶ Scoring: Calculate the binding affinity using scoring functions.
- Example: A molecule like ibuprofen can be docked to the active site of COX-2 protein to predict how well it will bind and reduce inflammation.

Introduction to Molecular Docking:

- Scoring Functions: Quantify ligand-receptor interaction.
 - ▶ Takes a pose as input and returns a number indicating the likelihood that the pose represents a favorable binding interaction;
 - ▶ Distinguish the near native poses from the nonnative poses of a particular ligand molecule
 - ▶ The predicted Protein-ligand binding affinities should correlate with the experimentally derived values
 - ▶ The scoring function should have a low computational cost, making it applicable to a large set of ligand molecules
 - ▶ low (negative) energy indicates a stable system and thus a likely binding interaction
- Empirical:
 - Fits experimental data to mathematical models
 - Based on energy terms (e.g., van der Waals forces, hydrogen bonds)
 - e.g., GlideScore, AutoDock's energy-based scoring
- Knowledge-based:
 - Derived from statistical analysis of existing protein-ligand complexes.
- Force-field-based: Use molecular mechanics equations (eg., AMBER, CHARMM).

Overview of Molecular Modelling:

- Molecular modeling is like designing molecules on a computer. It allows us to see, build, and simulate molecules, such as proteins or drugs.
- Imagine using software to build a LEGO model of molecules like proteins or drugs.
- Molecular modelling includes methods to predict and visualize the behavior of molecules. This involves:
 - ▶ Molecule Visualization: Creating 3D structures using software.
 - ▶ Force Field: A set of equations to model the interactions between atoms.
 - ▶ Quantum Chemistry: Describes molecules using the principles of quantum mechanics.
 - ▶ Energy Minimization: Finding the most stable structure (lowest energy state) of a molecule.
- Key terms:
 - ▶ Local minimum: Temporary stable structure.
 - ▶ Global minimum: The most stable, lowest-energy structure.
 - ▶ Saddle Point: Transition states between two local minima.
- Grid Search is used to systematically identify stable configurations.

Overview of Molecular Modelling:

- Molecular modeling involves simulating molecular structures, their physical properties, and behavior using computational tools and mathematical methods. It is widely used in drug discovery, protein structure prediction, and materials science.
- Molecular modelling relies on:
 - ▶ Schrödinger Equation:
 - Fundamental in quantum chemistry to predict electronic properties.
 - Solves for the electronic structure of molecules to predict reactivity and stability.
 - ▶ Energy Functions:
 - Force Fields: Equations used to simulate molecular behavior.
 - Stretching bonds, bending angles, and rotating torsions
 - To determine molecular stability:

$$E_{\text{total}} = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{non-bonded}}$$

- Example: Using Gaussian software, one can optimize the structure of benzene and predict its electronic energy levels.

Overview of Molecular Mechanics:

- Molecular mechanics (MM) uses classical mechanics to model molecular systems and calculate energy using force fields. Molecular mechanics treats atoms like balls and bonds like springs.
- Molecular mechanics models molecules like a spring-based system, where bonds behave like tiny springs. Atoms move like balls connected by springs.
- Molecular mechanics uses physics-based models to describe the behavior of molecules:
 - ▶ Force Fields: Define how bonds (stretching, bending) and interactions behave.
 - ▶ Non-Bonded Interactions:
 - Van der Waals forces: Weak attractions or repulsions.
 - Coulomb interactions: Electrostatic forces between charged atoms.
 - ▶ Derivative Methods:
 - Steepest Descent: Moves quickly downhill on an energy curve.
 - Conjugate Gradient: Optimizes movement to reach energy minima faster.
 - Newton-Raphson: Uses curvature of energy surfaces for precise minimization.
- Applications:
 - ▶ Protein-ligand interaction studies.
 - ▶ Predicting material properties.
 - ▶ Drug design and lead optimization.

Molecular Modelling:

- Molecular Representation:
 - ▶ Ball and Stick Model: Atoms represented as spheres and bonds as sticks.
 - ▶ Space-Filling Model: Represents the van der Waals radii of atoms.
 - ▶ Wireframe Model: Bonds are represented as lines.
- Force Field:
- A mathematical function describing the potential energy of a molecular system based on atomic positions. It includes terms for:
 - ▶ Bond stretching: Harmonic potential describing changes in bond lengths.
 - ▶ Angle bending: Potential for bond angles.
 - ▶ Dihedral torsions: Energy related to the rotation of bonds.
 - ▶ Non-bonded interactions:
 - Van der Waals forces.
 - Electrostatic interactions (Coulomb potential).

Molecular Modelling:

- Quantum Chemistry:
 - ▶ Based on Schrödinger's equation to describe electrons' behavior.
 - ▶ Used for small molecules and accurate binding energy calculations.
 - ▶ Methods: Ab Initio (e.g., Hartree-Fock) and Density Functional Theory (DFT).
- Energy Minimization:
 - ▶ Goal: To find the global energy minimum where the molecular system is most stable.
 - ▶ Saddle Points: Local minima that are not the most stable configuration.
 - ▶ Methods:
 - Steepest Descent: Moves atoms iteratively to minimize energy quickly but may stop at a local minimum.
 - Conjugate Gradient: Optimizes faster than steepest descent.
 - Newton-Raphson: Accurate but computationally expensive.
- Local vs Global Minima:
 - ▶ Local Minima: Low-energy configurations that may not represent the true stable form.
 - ▶ Global Minimum: The most stable energy state.

SBDD:

- SBDD leverages the 3D structure of the target protein to design small molecules (ligands) that can interact with the target to achieve a therapeutic effect.
- The primary goal is to identify compounds that bind strongly and specifically to the active site or another relevant site on the protein.
- Step 1: Protein preparation
- Step 2: Active site identification
- Step 3: Ligand Preparation
- Step 4: Molecular Docking
- Step 5: Scoring & Ranking
- Step 6: Molecular Dynamics Simulation
- Step 7: ADMET Prediction
- Step 6: Iterative Refinements
- In practice, SBDD and LBDD are often used together:
 - ▶ LBDD generates a library of promising ligands.
 - ▶ SBDD evaluates their binding modes in the target protein.

SBDD Workflow:

- Target Selection:
 - ▶ Identify the protein involved in the disease pathway.
 - ▶ Obtain its 3D structure from databases like RCSB PDB.
- Binding Site Identification:
 - ▶ Use tools like CASTp or PrankWeb.
- Ligand Preparation:
 - ▶ Generate ligand structures using PubChem or ChemSketch.
 - ▶ Optimize ligand geometries using QM (e.g., Gaussian).
- Docking:
 - ▶ Perform docking using AutoDock or Vina.
- Molecular Dynamics:
 - ▶ Simulate the docked complex using MD tools like GROMACS.
- Validation:
 - ▶ Analyze MD trajectories for stability.
 - ▶ Compare binding energies.

SBDD Workflow:

- Validation ensures the accuracy and reliability of docking simulations. Common methods include:
 - ▶ Re-docking: Testing the ability to reproduce a known binding pose.
 - ▶ Cross-validation: Using different scoring functions and datasets.
 - ▶ Experimental Validation: Comparing docking results with experimental data.

History:

- ❑ Pre-1970s: Drug discovery was largely experimental, involving trial-and-error testing of natural products and synthetic compounds.
- ❑ 1970s-1980s: The advent of molecular biology and structural techniques like X-ray crystallography enabled the rise of SBDD.
- ❑ 1990s: Computational advancements, including QSAR and pharmacophore modeling, made LBDD more accessible.
- ❑ 2000s-Present: Integration of artificial intelligence (AI) and machine learning (ML) has revolutionized both SBDD and LBDD, enabling faster and more accurate predictions

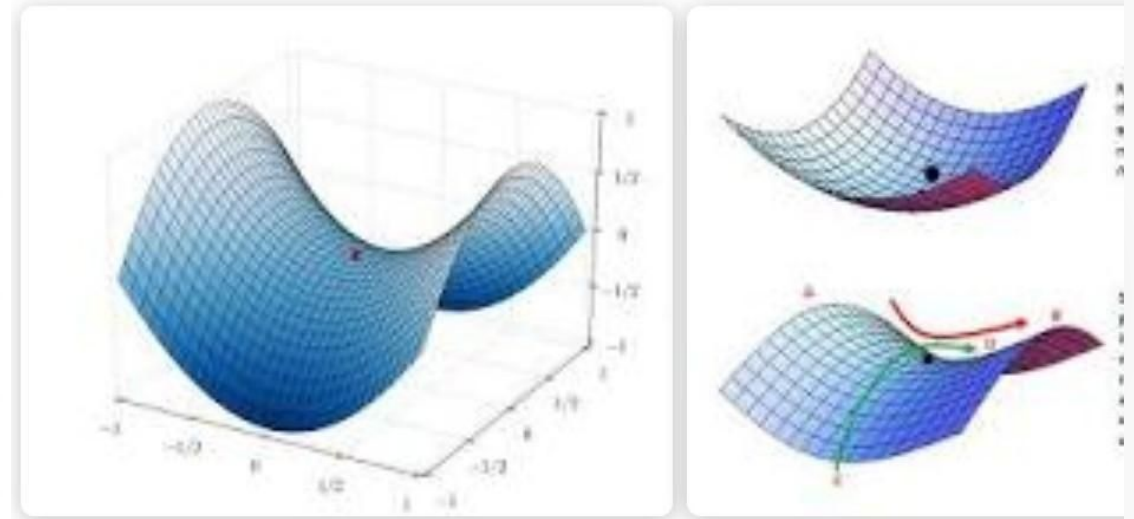
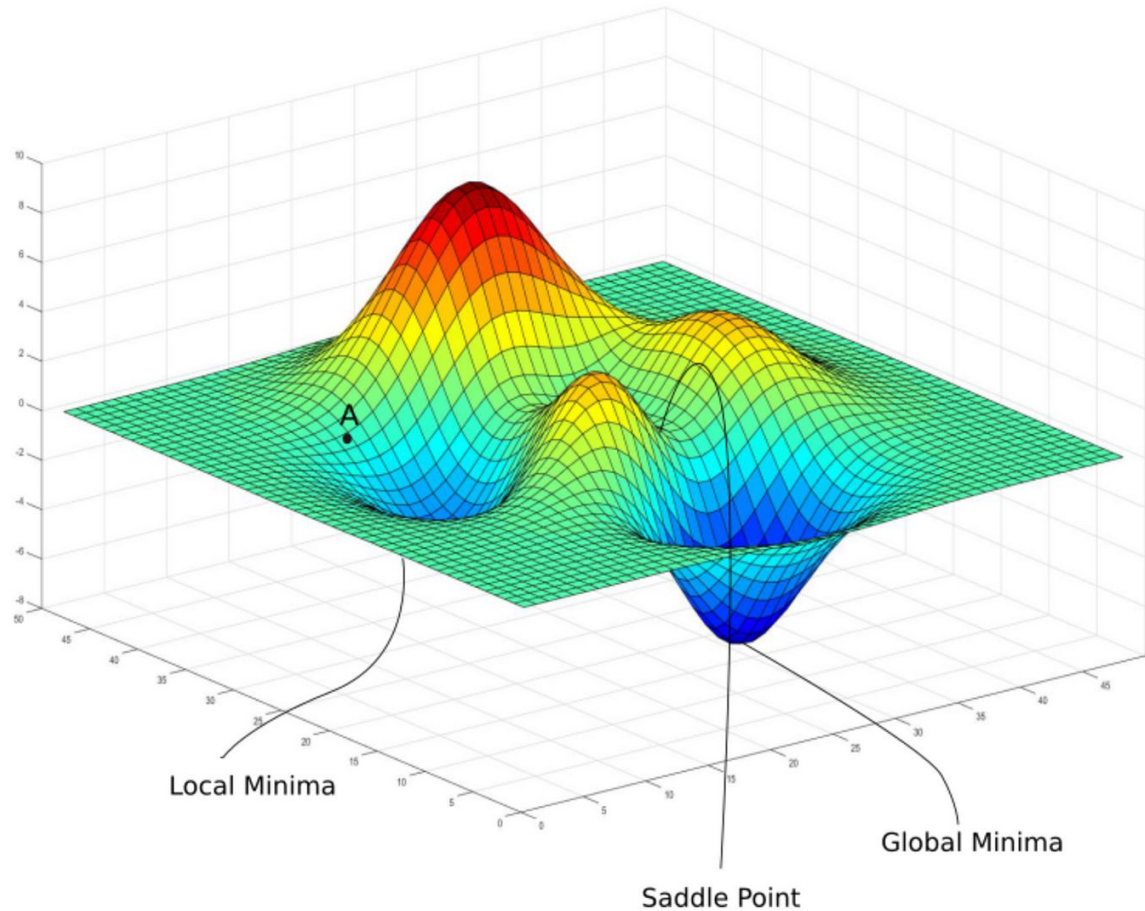
Schrödinger Equation:

- **What is the Schrödinger Equation?**
- Think of the Schrödinger equation as the master formula for quantum mechanics. It's like a GPS that tells electrons:
 - ▶ Where they're likely to be.
 - ▶ What energy they have.
- It looks like this (don't worry about solving it):
- $H\psi = E\psi$
- Here's what this means:
- H: The Hamiltonian operator, which represents the total energy of the system (kinetic + potential energy).
- ψ : The wavefunction, describing where the electron might be and its behavior.
- E: The energy of the system.
- The Schrödinger equation is like a magical crystal ball for molecules. It predicts where electrons are hiding and how much energy they have, helping scientists understand everything from chemical reactions to light absorption.
- The Schrödinger equation gives us:
 - ▶ Molecular Orbitals: Tells us where electrons hang out in a molecule.
 - ▶ Bond Strength: Predicts how strong chemical bonds are.
 - ▶ Spectroscopy: Helps explain how molecules absorb light.

Potential Energy Functions (Mapping the Molecular Landscape):

- Imagine a mountain range with peaks and valleys. Molecules, like hikers, are trying to find the lowest point (the most stable state).
- The potential energy function is a mathematical map of this landscape.
- Every position of a molecule corresponds to a specific energy level.
- The molecule “moves” by changing its shape (bond lengths, angles, etc.) to lower its energy.
- A **Potential Energy Surface (PES)** is a 3D plot that shows how a molecule’s energy changes as its shape changes.
- Scientists use it to predict:
 - ▶ Reaction pathways.
 - ▶ Transition states.
 - ▶ Stable conformations.
- Local Minima:
 - ▶ The energy well corresponds to a stable, but not necessarily optimal, structure.
 - ▶ Example: A protein might fold into a metastable state that is not the most efficient folding pathway.
- Global Minimum:
 - ▶ The lowest energy point in the landscape, representing the most stable conformation.
 - ▶ Example: The native structure of a protein corresponds to the global minimum in its folding landscape.
- Saddle Points:
 - ▶ These are transition states, like mountain passes. Molecules briefly pass through these states during chemical reactions.

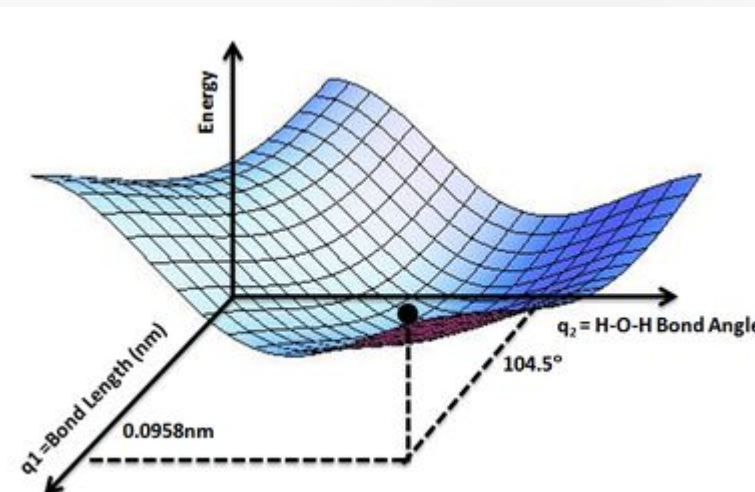
Local and Global Minima and Saddle point:



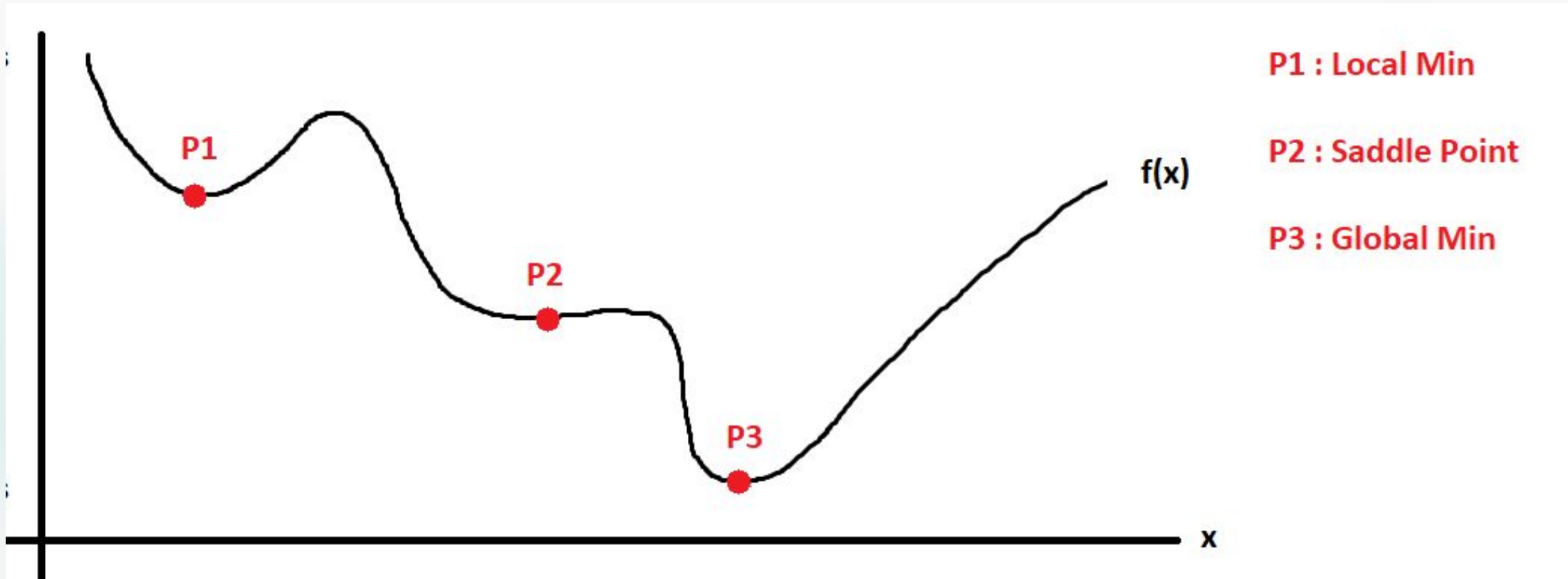
In mathematics, a **saddle point** or minimax **point** is a **point** on the surface of the graph of a function where the slopes (derivatives) in orthogonal directions are all zero (a critical **point**), but which is not a local extremum of the function.

Local and Global Minima and Saddle point:

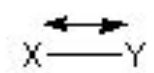
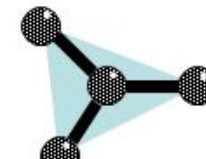
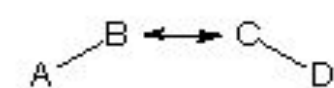
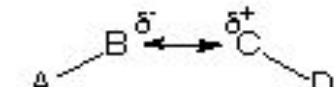
- Force Field: Rules that describe how atoms interact (like springs, angles, and magnets). Used in molecular simulations like AMBER and CHARMM.
- Quantum Chemistry: Explains how electrons behave as waves and particles, forming bonds and dictating molecular properties.
- Schrödinger Equation: The formula at the core of quantum mechanics, predicting where electrons are and how much energy they have.
- Potential Energy Function: A mathematical map of a molecule's energy landscape, showing stable states (valleys) and reaction pathways (saddle points).



Local and Global Minima and Saddle point:



Overview of Molecular Mechanics:

Bonded		$\sum K_b(b - b_0)^2$		$\sum K_\theta(\theta - \theta_0)^2$		$\sum V_n[1 - \cos(n\phi - \phi_n^0)]$	V_{str}		bond stretching
		$\sum K_\chi\chi^2$		$\sum K_{b\theta}(b - b_0)^2(\theta - \theta_0)^2$			V_{bend}		angle bending
							V_{tor}		torsional bending
							V_{oop}		out of plane bending
							V_{cross}		stretch/bend cross terms
Nonbonded	Electrostatic $\sum \frac{q_i q_j}{\epsilon r_{ij}}$						V_{vdW}		van der Waals nonbonded interactions
	Repulsion-dispersion $\sum \frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^6}$						V_{ES}		electrostatic interactions

Energy Minimization:

- Energy minimization relies on understanding and manipulating the energy gradient, which represents the rate of change of potential energy with respect to atomic positions. Derivative methods are classified into 1st-order and 2nd-order methods based on how the energy gradient is used in the minimization algorithm.
- Imagine a marble rolling on a bumpy hill. The marble wants to settle in the lowest possible point because that's the most stable position (lowest energy). In molecules, atoms also want to find their most stable arrangement (minimum energy) because nature prefers stability.
- But molecules don't magically fall into this perfect arrangement – we need to help them move downhill using mathematical methods. That's where energy minimization techniques come in. These are like guides that tell the molecule: “Go this way, a little more, a little less, until you're comfortable.”
- **Derivative Methods**
 - ▶ To help the marble (or molecule) move downhill, we need to figure out:
 - Where it's currently standing (current position).
 - Which direction leads downhill (gradient or slope).
 - How steep the slope is (gradient magnitude or curvature).
 - ▶ Derivative methods use math to calculate these slopes and decide how to move the molecule toward the lowest energy point.

Energy Minimization:

- Energy minimization is the process of finding the most stable (low-energy) conformation of a molecule. Molecules naturally tend to adopt conformations with the lowest potential energy, which corresponds to the most stable structure. This process is critical in molecular modeling to:
 - ▶ **Refine molecular geometries** before simulations.
 - ▶ **Remove steric clashes** and unrealistic conformations.
 - ▶ **Prepare molecules** for docking or dynamics simulations.

Energy Landscape

- The energy landscape represents the potential energy of a molecule as a function of its atomic positions (degrees of freedom). Visualizing this landscape helps understand molecular stability.
 1. Axes of the Landscape:
 - ▶ X-axis: Conformational space (atomic coordinates or bond angles).
 - ▶ Y-axis: Potential energy.
 2. Features of the Landscape:
 - ▶ Local Minima: Represent relatively stable conformations where the molecule is trapped in a low-energy state.
 - ▶ Global Minimum: Represents the most stable conformation with the lowest energy.
 - ▶ Saddle Points: High-energy points between minima, representing transition states.
- Imagine a hilly terrain, where valleys represent minima. A molecule in a local valley (local minimum) may require "energy" to climb over hills (saddle points) and reach the lowest valley (global minimum).

Steps in Energy Minimization:

1. Initial Structure Preparation:
 - a. Load the molecular structure (from a PDB or SMILES file).
 - b. Assign atomic charges and force field parameters.
2. Energy Calculation:
 - a. Compute the potential energy using a force field.
 - b. Identify steric clashes or high-energy regions.
3. Iterative Minimization:
 - a. Apply algorithms to adjust atomic positions iteratively to reduce energy.
 - b. Examples of algorithms:
 - i. Steepest Descent: Moves atoms downhill along the energy gradient. Simple but slow near the minimum.
 - ii. Conjugate Gradient: Uses past gradients to speed up convergence. More efficient for large systems.
4. Convergence Check:
 - a. The process stops when the energy gradient becomes negligible (below a threshold).

Comparison Of Methods

□ Visualization of Energy Landscape

- ▶ Steepest Descent: Moves straight downhill in small steps but can zigzag near the minimum.
- ▶ Conjugate Gradient: Adjusts the path to avoid zigzagging, leading to a smoother and faster descent.
- ▶ Newton-Raphson: Takes precise steps directly toward the minimum (if the Hessian is accurate).

□ Example Visualization:

□ Imagine a rolling ball on a bumpy surface:

- ▶ Steepest Descent: The ball follows the steepest path downhill but may get stuck in small pits (local minima).
- ▶ Conjugate Gradient: The ball remembers its previous path, allowing it to glide smoothly over bumps.
- ▶ Newton-Raphson: The ball "calculates" the ideal direction and step size to quickly reach the lowest point.

□

Comparison of Methods

Feature	Steepest Descent	Conjugate Gradient	Newton-Raphson
Gradient Usage	1st-order	1st-order + previous step	1st- and 2nd-order (Hessian)
Convergence Speed	Slow	Moderate	Fast (near minimum)
Complexity	Simple	Moderate	Complex
Computational Cost	Low	Moderate	High
Use Case	Initial minimization	Larger systems	Near-minimum refinement

1st-Order Derivative Methods (Simple Slope Methods):

- These methods use the first derivative of the potential energy function, also known as the gradient, to guide the optimization process.
- **Steepest Descent Method – The Beginner’s Approach**
- Principle: Moves atoms in the direction of the negative gradient (steepest descent direction), which points downhill on the energy landscape.
- Think about standing on a mountain with a steep slope. You want to get to the bottom, so you take a step directly downhill (in the steepest direction). You repeat this process, step by step, until you can’t go any lower.
- How it works for molecules:
 - ▶ We calculate the steepest energy slope for the molecule, move the atoms slightly, and repeat. It’s like saying, “Okay, this way is down. Let’s move a bit and check again.”
- Pros:
 - ▶ Super simple and works well when the molecule is far from stable. Think of it as a good beginner tool.
- Cons:
 - ▶ It’s slow. The closer you get to the bottom, the smaller the steps become. It’s like taking baby steps when you’re almost at the valley.
- Example:
 - ▶ If you’ve got a protein and its atoms are in weird positions (like atoms crashing into each other), steepest descent helps untangle them gently.

1st-Order Derivative Methods (Simple Slope Methods):

- Refer notes for Steps and Derivative Methods in Energy Minimization

Techniques in Molecular Modelling:

- ❑ **Homology Modelling:**
- ❑ Homology modeling is like copying homework from a friend who solved a similar problem. You make some adjustments to fit your situation but rely heavily on their work.
- ❑ Predicts protein structures based on sequence alignment with a known template.
- ❑ **Applications:**
 - ▶ When experimental structures (e.g., X-ray crystallography or NMR) are not available.
 - ▶ Works best if the template shares >30% sequence identity with the target.
- ❑ Proteins are made of amino acid sequences, and their structure determines their function. If we don't have a structure for a protein, but we have a known structure of a similar protein (template), we can use homology modeling to predict the unknown structure.

Techniques in Molecular Modelling:

- ❑ **Working of Homology Modelling:**
- ❑ **Template Selection:**
 - ▶ Search a protein database (like PDB) for a protein with a known 3D structure that has a similar amino acid sequence.
 - ▶ Example: If you have a protein from a fish, you might use a similar protein structure from a frog as the template.
- ❑ **Sequence Alignment:**
 - ▶ Align the sequence of your unknown protein (target) with the template. This step is like matching similar sentences in two languages.
- ❑ **Model Building:**
 - ▶ Replace the amino acid sequence of the template with the target's sequence while maintaining the backbone structure.
- ❑ **Refinement:**
 - ▶ Adjust side chains and optimize the structure to fit the chemical properties of the target protein.
- ❑ **Validation:**
 - ▶ Use tools (e.g., Ramachandran plot) to check if the predicted structure is realistic.

Techniques in Molecular Modelling:

- **Ab Initio Methods:**
- Ab initio is like trying to assemble a puzzle without looking at the picture on the box. You rely only on logic and how the pieces fit together.
- Ab initio (Latin for "from the beginning") methods attempt to predict a protein's structure from scratch, without relying on any existing template. Instead, they use physics and chemistry principles to simulate how the protein folds.
- Quantum mechanical calculations without empirical parameters.
- **Working of Ab initio methods rely on quantum mechanics and molecular dynamics:**
 - ▶ Energy Calculations:
 - Use the Schrödinger equation to calculate the energy of different possible structures.
 - ▶ Fold Prediction:
 - The protein folds into the shape with the lowest energy, so these methods simulate various folding pathways to find that stable state.
- **Challenges:**
 - ▶ Computationally Expensive:
 - Simulating all possible conformations for a protein can take a massive amount of computing power.
 - ▶ Accuracy:
 - Works well for small proteins (<100 amino acids) but struggles with larger ones.

Techniques in Molecular Modelling:

- ❑ **Molecular Dynamics Simulations:**
- ❑ Models the movement of molecules over time.
- ❑ Molecular dynamics simulations model the movement of molecules over time, like creating a movie of how a protein or molecule behaves in a solution.
- ❑ Applications of MD Simulations:
 - ▶ Study how drugs bind to proteins.
 - ▶ Understand protein folding and dynamics.
 - ▶ Predict the stability of biomolecular systems.
- ❑ MD simulations are like filming a slow-motion video of a dancer (protein) moving in a crowded room (water). Every step and twist is calculated based on physical forces acting on the dancer.

Techniques in Molecular Modelling:

- **Working of MD Simulations:**
- Starting Structure:
 - ▶ Begin with a protein's structure (from experiments or homology modeling).
- Environment Setup:
 - ▶ Place the protein in a simulation box with water molecules, ions, and other relevant conditions (e.g., temperature, pH).
- Apply Physics Laws:
 - ▶ Use Newton's laws of motion to calculate how each atom in the system moves under forces like:
 - Bond stretching.
 - Electrostatic interactions.
 - Van der Waals forces.
- Run the Simulation:
 - ▶ The program calculates these movements at femtosecond intervals (10^{-15} seconds) and tracks how the structure evolves over time.
- Analyze Results:
 - ▶ Extract information like:
 - Root Mean Square Deviation (RMSD): How much the structure changes over time.
 - Radius of Gyration: How compact the protein is.
 - Hydrogen Bonds: How many bonds are formed/stable

Techniques in Molecular Modelling:

Aspect	Homology Modeling	Ab Initio Methods	Molecular Dynamics
Purpose	Predict protein structure from a template	Predict structure from scratch	Simulate molecular motion over time
Input	Target sequence + template structure	Only the amino acid sequence	Protein structure, solvent, and conditions
Computational Cost	Moderate	Very high	High
Accuracy	Depends on template similarity	Better for small proteins	Accurate for motion but requires good input
Applications	Structure prediction	Structure prediction	Dynamics, stability, and interactions

Overview of Molecular Dynamics

Concept	Definition	Key Takeaway
Newton's Equations	Govern how atoms move by calculating their forces, accelerations, and updated positions.	Foundation for all MD simulations.
Equilibrium Point	The most stable configuration where forces balance, and energy is minimized.	Proteins naturally fluctuate near this point in simulations.
RDF (g(r))	Describes how particles are distributed around a reference particle in space.	Used to study molecular organization and solvent behavior.
Pair Correlation Function	Probability of finding a pair of particles separated by a certain distance.	A statistical tool to analyze atomic or molecular distributions.
MD Run Duration	Simulation time required to capture relevant molecular movements or biological processes.	Varies from nanoseconds to milliseconds depending on the system and research question.

References

- Stich et al., "Conjugate gradient minimization of the energy functional: A new method for electronic structure calculation", Physical Review B 39 (8), pp. 4997–5004, (1989)
<https://journals.aps.org/prb/abstract/10.1103/PhysRevB.39.4997>
- The Golden Age of Quantum Physics (1927):
<https://www.privatdozent.co/p/the-golden-age-of-quantum-physics-ff7>
- Andrew R Leach, Molecular Modelling - Principles and Applications:
https://drive.google.com/file/d/18Pq0D0xPqKv99_gySETLCl0OZnkcgLz/view?usp=sharing



Thank-You!